

Split Complex Lorentzian Dynamics

Part XV: Regular Central Lifts, Direction-Dependent Residual Rates, and the Joint-Regularity Gap

A formally verified continuation of the dependency audit

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Abstract

Part XIV closed the noncentral spatial problem much further than had been guaranteed in advance. Independently reconstructed axis systems generated the third spatial direction, the three infinitesimal generators closed, arbitrary-axis reference implementations were available, and the universal minimal-carrier type survived. The obstruction did not disappear. It migrated into the exact center

$$Z = \text{Center}(W) = \text{span}_{\mathbb{R}}\{1, S\}, \quad S^2 = -1,$$

where reference words retained an exact $\{+1, -1\}$ defect, unrestricted implementations differed by arbitrary elements of $Z^\times$, and no globally multiplicative implementer assignment existed. The remaining open structure was therefore no longer plausibly another spatial generator. It concerned continuity, rates, composition paths and central residual data.

Part XV attacks precisely that boundary. For one fixed unit axis n , every internal lift of the inherited one-parameter automorphism family factors uniquely as a central multiplicative residual times the already reconstructed reference implementation. Continuity then rigidly classifies the residual:

$$U_n(\theta) = e^{\alpha\theta}(\cos(\beta\theta)1 + \sin(\beta\theta)S)U_n^{\text{ref}}(\theta),$$

with unique real parameters α, β . Differentiability adds no further freedom: every continuous lift in this class is already differentiable. Thus the formerly arbitrary pointwise central factor has been reduced, along each one-parameter subgroup, to exactly two rates.

The finite mixed-composition problem is then formalized without quotienting the center. For an implementer choice U the discrepancy $c(g, h)$ is an invertible central element satisfying

$$U(g)U(h) = c(g, h)U(g \circ h),$$

and associativity forces the exact three-term relation

$$c(g, h)c(g \circ h, k) = c(h, k)c(g, h \circ k).$$

For regular per-axis lifts the defect separates into the accumulated continuous residual and the inherited reference-word sign defect. Only after this finite law is fixed is the system differentiated. The infinitesimal generator for direction n becomes

$$G(n) = \alpha(n)1 + \beta(n)S - \frac{1}{2}J(n),$$

with the central and spatial pieces uniquely separated. The spatial reparametrization rate, α and β are formally independent coordinates of the generator.

The official Task-XV verdict is that direction-dependent central data survive: for the formal predicate called a “regular multi-axis family”, every pair of real functions $\alpha(n), \beta(n)$ is

realized. This statement is correct for the definition that was proved. However, the definition imposes continuity only in the one-parameter variable θ for each fixed direction. It explicitly imposes no continuity or differentiability in n . Consequently the theorem does not yet answer the stronger question that motivated the task: joint regularity across spatial direction and transformation parameter. This is not a defect in the Lean theorem; it is a newly exposed dependency boundary that must be hardened before the result is interpreted as full continuous $3 + 1$ coherence.

A second hardening point is even sharper. Under the stronger requirement that equal visible automorphisms receive equal internal implementers, Task XV proves necessary conditions

$$\alpha(n) = 0, \quad \beta(n) \in \mathbb{Z} + \frac{1}{2}, \quad \beta(-n) = -\beta(n).$$

Sufficiency and existence are not proved because the exact coincidence relation among arbitrary parameter pairs (n, θ) representing the same automorphism is not yet classified. Moreover, if joint directional continuity were to force β to be continuous on the connected unit sphere, these necessary conditions would be in immediate tension: a continuous map into the discrete set $\mathbb{Z} + \frac{1}{2}$ is constant, while antipodal oddness would force that constant to vanish, which is impossible. This implication is a paper-level diagnostic, not a Task-XV Lean theorem. It identifies a very narrow next obstruction to test.

The heuristic comparison therefore also remains conditional. Task 1 Option C had two variable quantities constrained by one synchronization relation and was underdetermined. Task XV, under axiswise regularity only, yields two directional rates and no cross-direction relation; this is formally an even larger underdetermination. Whether the strengthened jointly regular, transformation-valued problem restores one relation, forces a discrete obstruction, or requires local/pathwise relational data is still open. The next task should close this entire boundary in one focused development: exact transformation coincidences, joint direction-parameter regularity, existence or no-go for transformation-valued lifts, the resulting regular central composition defect, and, if global closure fails, the minimal local repair. This is a compact hardening program rather than a broad new physical task.

Keywords: continuous lift; exact center; residual rate; mixed-axis composition; central defect; transformation-valued implementation; joint regularity; formal verification; Lean 4; conservation of difficulty.

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1 The boundary inherited from Part XIV

The logic of Part XIV was asymmetric in an important way. Almost every question about the visible spatial algebra closed positively: a second axis could be reconstructed without copying the first, the mixed commutator generated the third direction already present in W , the three infinitesimal generators closed, the four-real-dimensional reference implementer algebra closed, and every unit spatial direction admitted the same reduced carrier law. The negative answers appeared only when one asked for a globally coherent internal realization [7, 1].

The inherited ambient data are

$$W = \text{span}_{\mathbb{R}}\{1, A, B, C, P, Q, R, S\}, \quad Z = \text{Center}(W) = \text{span}_{\mathbb{R}}\{1, S\}, \quad S^2 = -1. \quad (1)$$

For each unit spatial direction n there is an inherited automorphism family $\Phi_n(\theta)$ and a normalized reference implementation

$$U_n^{\text{ref}}(\theta) = \cos(\theta/2)1 - \sin(\theta/2)J(n), \quad (2)$$

where the already reconstructed linear map J satisfies the spatial product laws of Part XIV.

Two facts from Part XIV define the new problem. First, two internal implementations of the same visible automorphism differ centrally. Second, no exact globally multiplicative selection exists. In particular, a closed visible path can leave the internal value -1 even though the visible automorphism has returned to id.

This is the appropriate Conservation-of-Difficulty boundary: the spatial carrier and spatial generators are no longer missing. The open datum concerns how internal representatives vary with parameters and how their central discrepancies compose.

Methodological principle 1 (Do not repair a central coherence defect with another spatial generator). Once the complete noncentral spatial implementation algebra has been proved closed and carrier-independent, a remaining central path defect must be analyzed in its own dependency class. Adding further noncentral spatial structure would solve a problem that has already been solved and would not address the proved obstruction.

2 Task-XV scope, source order and firewall

Task XV was deliberately narrower than Task XIV. It imports the Task-XIV reconstruction layer and the Task-XI continuous one-axis classification, then proceeds through nine modules in the order

$$\begin{aligned} \text{SafeBase} &\rightarrow \text{RegularLift} \rightarrow \text{DifferentiableLift} \rightarrow \text{MixedComposition} \\ &\rightarrow \text{Infinitesimal} \rightarrow \text{RateSeparation} \rightarrow \text{ResidualAudit} \\ &\rightarrow \text{OptionCAudit} \rightarrow \text{Task15}. \end{aligned}$$

No physical target language is used in the reconstruction layer. The formal firewall excludes particle names, physical spin dynamics, mass, energy, Hamiltonians, field equations, gauge or gravitational language, projective-representation terminology and the hole-tape heuristic. The Option-C comparison is imported only by the top module and contributes no theorem to the reconstruction chain [2].

This source order matters because the task asks three logically distinct questions:

1. what continuity alone forces on one-axis central residuals;
2. what finite mixed composition forces before differentiation;
3. what remains free after the exact infinitesimal generator is isolated.

The third question cannot legitimately be answered by assuming the expected exponential form at the beginning.

3 Exact factorization of an arbitrary axis lift

Fix a unit spatial direction n . Task XV defines an axis lift $U : \mathbb{R} \rightarrow W$ by three requirements: unit value at zero, the one-parameter group law, and implementation of the inherited $\Phi_n(\theta)$ by conjugation. No continuity is needed for the first structural theorem.

Theorem 1 (Unique central factorization). *For every axis lift U there exists a unique multiplicative central map $z : \mathbb{R} \rightarrow Z^\times$ such that*

$$U(\theta) = z(\theta)U_n^{\text{ref}}(\theta) \quad (3)$$

for every θ .

The formal residual is not introduced as a free ansatz. It is computed from the given lift and the already known inverse reference implementation. The exact center theorem from the earlier tasks then forces the residual into Z . This is the first important narrowing relative to Part XIV: the central discrepancy is not merely known to exist pointwise; its one-axis multiplicative structure is now explicit.

4 Continuity forces exactly two real rates

The first principal theorem of Task XV adds continuity of U in the parameter θ . The resulting classification is complete.

Theorem 2 (Continuous axis-lift classification). *Let n be a unit axis. A continuous axis lift is equivalent to the existence of unique $\alpha, \beta \in \mathbb{R}$ such that*

$$U(\theta) = z_{\alpha,\beta}(\theta)U_n^{\text{ref}}(\theta), \quad (4)$$

where

$$z_{\alpha,\beta}(\theta) = e^{\alpha\theta}(\cos(\beta\theta)1 + \sin(\beta\theta)S). \quad (5)$$

The significance is not the familiar form of equation (5); the significance is its position in the dependency graph. The form is derived from the exact central plane, multiplicativity and continuity. It is not inserted as a normalization or as a physical phase law.

The two real parameters play structurally different roles inside the center. The α term generates the real scaling direction 1, while the β term generates the $S^2 = -1$ direction. At this stage both are simply central rates.

4.1 Differentiability adds nothing

Task XV then strengthens regularity and proves that the continuous class is already differentiable.

Theorem 3 (No additional differentiable class). *Every continuous axis lift in theorem 2 is differentiable, and every differentiable axis lift lies in the same two-parameter family.*

Thus a potentially much larger path-parameter freedom disappears. Under the one-parameter group law, there is no arbitrary function $\alpha(\theta)$ or $\beta(\theta)$ left. The freedom is two real constants along each fixed one-parameter subgroup.

Status 1 (First Task-XV closure). For a fixed unit axis, continuity closes the central residual class completely: two and only two real rates survive. Differentiability does not further reduce the class.

This is already a movement toward the dynamical side in a precise but limited sense. What remains open is no longer the functional form along a fixed orbit; it is how the two rates assigned to different directions are related.

5 The finite central composition law

Task XV deliberately analyzes finite composition before differentiating. For an implementer choice u and a compatible inverse choice u_i , define the central discrepancy of two implementable transformations g, h by an explicit product. The formal theorems show

$$u(g)u(h) = c(g, h)u(g \circ h), \quad c(g, h) \in Z^\times. \quad (6)$$

Associativity of W alone then gives

$$c(g, h)c(g \circ h, k) = c(h, k)c(g, h \circ k). \quad (7)$$

The unit normalizations are also derived, and a central redefinition of the implementer choice changes c by the corresponding three-factor central expression.

No quotient is introduced. The discrepancy remains an actual element of the reconstructed center. This is methodologically important because Part XIV had shown that the central residual carries exactly the information lost by the visible spatial automorphism.

5.1 Reference defect and regular residual separate

For regular per-axis lifts, Task XV proves an exact word-factorization theorem. A mixed internal word separates into

accumulated continuous central residual $\times$ inherited reference word.

When two words implement the same visible transformation, the reference component contributes the already known $\{+1, -1\}$ defect, while the continuous residual contributes the product of the z factors. The two sources of central discrepancy are therefore not conflated.

Methodological principle 2 (Different defects must remain different). The discrete reference-word sign and the continuous $Z^\times$ residual are both central, but they arise from different dependency layers. They must not be identified merely because both multiply the same internal representative.

6 Infinitesimal central data

Only after the finite law is fixed does Task XV differentiate the regular lift at the identity. For a direction n the generator is

$$G(n) = C(n) - \frac{1}{2}J(n), \quad (8)$$

with

$$C(n) = \alpha(n)1 + \beta(n)S \in Z. \quad (9)$$

Because $1, S$ are an exact real basis of the center, the coefficients $\alpha(n), \beta(n)$ are unique.

This gives a particularly clean decomposition:

$$\boxed{G(n) = \underbrace{\alpha(n)1 + \beta(n)S}_{\text{central residual}} - \underbrace{\frac{1}{2}J(n)}_{\text{spatial reference generator}}}. \quad (10)$$

The decomposition is formal and does not require a physical meaning for either term.

7 The official Task-XV verdict

Task XV introduces a predicate called *regular multi-axis family*. Its exact definition is crucial:

$$\text{Reg}(U) \quad :\Longleftrightarrow \quad \text{for every unit } n, \theta \mapsto U(n, \theta) \text{ is a continuous axis lift.} \quad (11)$$

The source comment explicitly states that no relation between different directions is imposed.

For this predicate Aristotle proves a complete theorem.

Theorem 4 (Arbitrary directional rate functions). *For every pair of real functions*

$$\alpha, \beta : \{n : h(n, n) = 1\} \rightarrow \mathbb{R},$$

the family

$$U(n, \theta) = z_{\alpha(n), \beta(n)}(\theta) U_n^{\text{ref}}(\theta) \quad (12)$$

is a regular family in the sense of equation (11), and its recovered rates are exactly $\alpha(n), \beta(n)$.

Consequently two different directions can carry completely unrelated rate pairs. Constants are too restrictive; arbitrary functions of direction are exactly the datum classified by this definition. At the same time there is no additional free function of θ : the one-axis group law and continuity already removed that freedom.

The formal Task-XV verdict is therefore

DIRECTION-DEPENDENT CENTRAL STATE DATA SURVIVE.

The word “state” here is only the task’s neutral label for the surviving internal datum. No physical state space has been reconstructed.

7.1 What this theorem does and does not say

The theorem is strong and exact, but its quantifiers must be read literally. It proves

$$\forall n : \quad \text{regularity in } \theta.$$

It does *not* prove

$$\text{joint regularity in } (n, \theta).$$

Therefore the statement “continuous 3 + 1 coherence imposes no relation between directions” would be stronger than the formal theorem. The theorem instead says:

If every axis is required to carry its own continuous one-parameter lift, and no cross-axis regularity is included in the definition, then the two rates may be assigned independently from axis to axis.

This distinction is the main hardening obligation exported by Task XV.

Methodological caveat 1 (Axiswise regularity is not joint multi-axis regularity). *The freedom of arbitrary, possibly discontinuous, functions $\alpha(n), \beta(n)$ is partly a consequence of the formal predicate: it deliberately imposes no continuity in n . The resulting theorem is correct, but it cannot settle whether a jointly continuous family permits the same freedom.*

8 Spatial rate and central rates are independent

Part XIV had separately shown that once the spatial generator assignment is required to be linear over all directions, isotropy forces one common normalization of the spatial rate. Task XV checks whether that result has any hidden consequence for α or β .

Introduce a spatial reparametrization rate λ . The generator becomes

$$G_{\lambda,\alpha,\beta}(n) = \alpha(n)1 + \beta(n)S - \frac{\lambda}{2}J(n). \quad (13)$$

The formal rate-separation theorem shows that λ , α and β are independently recoverable from the generator. Fixing one common λ leaves arbitrary central rate functions; fixing the central rates leaves λ free.

Thus

$$\boxed{\text{spatial rate coherence} \not\Rightarrow \text{central rate coherence.}} \quad (14)$$

This negative result matters because otherwise the Task-XIV spatial closure could have been used accidentally as a hidden normalization of the Task-XV central sector.

9 The stronger transformation-valued requirement

The task then asks a stricter question. A family is called transformation-valued when two parameter pairs that induce the same inherited visible automorphism are assigned the same internal implementer:

$$\Phi_n(\theta) = \Phi_m(\varphi) \implies U(n, \theta) = U(m, \varphi). \quad (15)$$

This is strictly stronger than axiswise regularity.

Task XV obtains three necessary consequences.

Theorem 5 (Necessary transformation-valued constraints). *For a regular transformation-valued family and every unit direction n ,*

$$\alpha(n) = 0, \quad (16)$$

$$\beta(n) \in \mathbb{Z} + \frac{1}{2}, \quad (17)$$

$$\beta(-n) = -\beta(n). \quad (18)$$

The corresponding antipodal relation for α is automatic from equation (16).

This is a major reduction of the continuous two-parameter central freedom. The real scaling rate is eliminated, while the second rate is forced into a discrete half-odd-integer set and must change sign under axis reversal.

It is essential, however, not to overstate the theorem.

Proof boundary 1 (Necessity is proved; sufficiency is not). *Task XV does not prove that a transformation-valued regular family satisfying equations (16) to (18) exists. It also does not prove that these conditions are sufficient. A sufficiency theorem requires the exact classification of all coincidences*

$$\Phi_n(\theta) = \Phi_m(\varphi),$$

which is explicitly listed as unresolved in the Task-XV audit.

Therefore phrases such as “the stronger condition leaves exactly one discrete function” must be read as a necessary-data statement, not yet as a complete existence classification.

10 Why Task XV must be hardened before physical continuation

The formal Task-XV layer is a mathematical pass: its theorem statements are proved and the audit records successful builds. The research interpretation, however, exports two principal obligations that cannot be skipped if the purpose is to locate the next forced structure rather than merely to accumulate correct lemmas.

10.1 Hardening point 1: regularity in direction

The intended phrase “continuous multi-axis coherence” naturally suggests regularity as n changes. The implemented predicate does not contain this requirement. Consequently the arbitrary-function theorem theorem 4 is not a negative answer to joint regularity; it is a classification of a weaker axiswise notion.

The next task must therefore introduce a separate stronger predicate, for example joint continuity of

$$(n, \theta) \longmapsto U(n, \theta)$$

on the unit-axis set times $\mathbb{R}$, while preserving the existing axiswise group law. Crucially, it should then *derive*, not assume, what regularity this imposes on the recovered coefficient functions $\alpha(n), \beta(n)$.

10.2 Hardening point 2: exact transformation coincidences

The transformation-valued condition quantifies over equality of visible automorphisms. To decide existence, one needs to know exactly when

$$\Phi_n(\theta) = \Phi_m(\varphi).$$

Task XV uses selected coincidences such as full turns and antipodal relations to derive necessary conditions, but it does not classify all coincidences. This exact relation is therefore the missing algebraic input for a necessity-and-sufficiency theorem.

This is narrower than a complete presentation of every spatial word relation. A full word problem is not required merely to settle whether the parameterized family descends to a well-defined function of the transformation.

10.3 Hardening point 3: the global-existence question

Once directional regularity and exact transformation coincidences are known, the existence problem must be closed with a theorem:

either construct a jointly regular transformation-valued family, or prove that none exists.

Leaving only necessary constraints would again move the decisive obstruction outside the verified boundary.

10.4 Hardening point 4: regularity of the central composition defect

If U is strengthened to be jointly regular, the central discrepancy $c(g, h)$ should be re-audited under that regularity. The static Task-XV theorem permits all of $Z^\times$. It remains to determine whether joint regularity restricts the continuously varying part, the discrete sign part, or their interaction.

This is directly relevant to the Part-XIV path memory and should be included in the same hardening task rather than postponed to an unrelated future branch.

11 A sharp diagnostic implication not yet formalized

The Task-XV theorems already place the next test in a very narrow corridor. Suppose a future hardening theorem establishes that a transformation-valued family is jointly continuous in direction and therefore that β is continuous on the unit sphere. Combine that with the already proved necessary conditions

$$\beta(n) \in \mathbb{Z} + \frac{1}{2}, \quad \beta(-n) = -\beta(n).$$

The set $\mathbb{Z} + \frac{1}{2}$ is discrete, while the real unit sphere is connected. Hence every continuous map from the sphere into this discrete set is constant. Antipodal oddness would then give

$$b = -b,$$

so $b = 0$, contradicting $0 \notin \mathbb{Z} + \frac{1}{2}$.

Proposition 1 (Conditional global no-go diagnostic). *If joint directional regularity forces β to be continuous, and if the Task-XV transformation-valued necessary conditions remain applicable globally, then no global jointly continuous transformation-valued family can exist.*

Proof boundary 2 (Status of proposition 1). *This proposition is a mathematical diagnostic used to design the next formal task. It is not one of the 37 exposed Task-XV Lean theorems. The next task should formalize the relevant implication directly, preferably from the intrinsic axis parametrization rather than by importing a conventional spin-cover argument.*

This potential no-go is especially significant for the Conservation-of-Difficulty methodology. The stronger coherence requirement would not merely choose a preferred α, β pair. It could show that no single global representative exists at all, forcing a more local or relational structure.

12 Return to the Option-C heuristic

The formal reconstruction contains no hole-tape terminology. After fixing the proof boundary, however, the earlier heuristic can be used as a diagnostic comparison [6].

In Task 1 Option C, marker structure and transport rate were both permitted to vary. Schematically the synchronization condition had the form

$$\text{density variable} \times \text{transport factor} \times \text{kinematic factor} = 1. \quad (19)$$

Thus there were two variable quantities but only one independent relation. The synchronization condition alone could not decide how much of the change belonged to the marker scale and how much to the transport rate.

The later heuristic dictionary, never used in Lean, reads

Formal central datum	Heuristic reading only
$e^{\alpha\theta}$	common hole-density / local marker-scale change
$e^{\beta S\theta}$	common band-transport / internal progression rate
$\mp \frac{1}{2}S$	rigid relative half-angle contribution
central path defect	internal memory after visible closure

Under the actual Task-XV predicate equation (11), the structural comparison is not Option C's "two quantities, one relation". It is

two directional rate functions + zero cross-direction relations.

That is an even larger underdetermination. The formal late comparison therefore correctly says that the Option-C dependency count does not reappear under axiswise regularity.

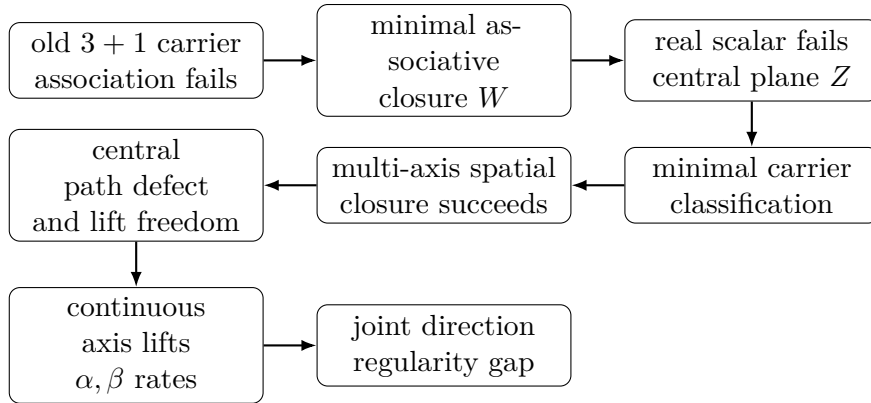
But that is not the end of the comparison. The stronger transformation-valued condition already introduces nontrivial relations, and joint directional continuity has not yet been tested. Therefore the scientifically relevant comparison must be postponed until the strengthened problem is closed. The possible outcomes are now much sharper:

1. joint regularity supplies exactly the missing relation and reduces the two rates;
2. joint regularity produces a discrete global obstruction rather than a relation between two continuous rates;
3. no global representative exists and the missing datum becomes local/pathwise relational information;
4. the heuristic analogy fails even after hardening.

Only after one of these is formally established should the heuristic words “density” and “transport” be revisited.

13 Conservation of Difficulty: what has moved, and what has not

The sequence from Parts VI–XV now shows a repeated pattern. Each apparently missing structure was either forced by a closure obstruction or removed from the list of candidates:



The unresolved difficulty has become more dynamical in mathematical type, but not yet in physical meaning. It concerns rates, continuous dependence, equality of different parameterizations, composition paths and internal residuals. No physical time variable, Hamiltonian, interaction field or equation of motion has been introduced.

Task XV also sharpens what *does not* constitute the missing dynamics. Along a fixed one-parameter subgroup, arbitrary time-dependent central functions are excluded by the group law and continuity. The surviving freedom is encoded in rates assigned to directions. Therefore a later physical dynamics, if it appears, cannot simply be identified with “let α and β be arbitrary functions of θ ”. The present algebra forbids that within the current one-parameter structure.

Methodological principle 3 (The difficulty should be hardened where it actually sits). A theorem that leaves arbitrary directional functions under a definition with no directional regularity should not be interpreted as evidence for arbitrary physical state dynamics. The next obligation is to strengthen the coherence predicate and determine whether the freedom survives that stronger test.

14 Formal verification and provenance

The supplied Aristotle archive for Task XV has SHA-256

a5e76f12d799d61541427f02fc7fa6b83025c36657e56833570940f772204688.

The pinned environment reported by the artifact is Lean v4.28.0 and mathlib commit

8f9d9cff6bd728b17a24e163c9402775d9e6a365.

The audit records successful builds of the Task-XV target and the complete project. Mechanical scans report no `sorry`, `admit`, custom axiom declaration or `@[implemented_by]` in the Task-XV layer. All 37 principal theorems exposed by `Task15.lean` report exactly

`[propext, Classical.choice, Quot.sound]`.

The formal theorem package is therefore internally closed within the stated definitions [2, 5, 8].

There is one minor documentation inconsistency worth recording because provenance itself is part of the experiment. The new `ARISTOTLE_SUMMARY.md` visibly contains a Task-XV entry, while that entry ends with the inherited sentence that the summary “was not modified”. The theorem files, Task-XV audit and README agree on the mathematical content; the inconsistency is confined to that prose sentence. It has no mathematical effect, but it illustrates why documentation claims must be checked against the delivered artifact rather than copied mechanically.

Proof boundary 3 (Verification boundary for this paper). *The delivered source tree, Task-XV audit, README, cumulative summary and principal Lean modules were inspected for this paper. The successful builds and axiom outputs are reported from the supplied Aristotle artifact; the full 8,000-plus-job Lean build was not independently rerun in the paper-generation environment.*

15 Late conventional comparison

Only after the intrinsic reconstruction is fixed may one compare it with standard mathematical language. The Part-XIV reference implementation already had the familiar double-valued spatial spinorial behavior [3, 4]. Part XV does not add a new standard spin representation. Instead it classifies the central freedom multiplying that reference action and exposes the exact conditions under which an internal representative would have to depend only on the visible transformation.

The finite identity equation (7) is the familiar algebraic consistency relation associated with a central composition defect, but no such conventional terminology is used in the reconstruction files. Likewise, the possible global obstruction suggested by proposition 1 should not be imported as a known topological answer. The experiment is specifically designed to see whether the obstruction can be reconstructed from the inherited algebra, the axis parameter space and regularity requirements themselves.

No physical spin observable, fermion, Hilbert-space norm, probability rule, mass, charge, full Lorentz-spin representation or field equation follows from Task XV.

16 What must be closed next

Task XV is not a “half proof”: every theorem it states is closed. But a dependency audit is not complete merely because the current theorem package builds. The task has revealed that the notion of regular multi-axis family used for its main verdict is weaker than the intended joint-coherence question, and it has left the transformation-valued existence problem at necessity only. Those are now the principal mathematical obligations.

At the same time, not every open item in the audit deserves equal priority. A full presentation of the spatial word problem is broader than needed for the present closure test, and a survey of measurable, bounded, smooth and analytic regularity classes would multiply branches without first resolving the decisive continuous case. The next task should therefore close the following compact set.

16.1 A focused Task XVI with six work packages

A neutral title is:

Task XVI – Joint Direction-Parameter Regularity and Transformation-Valued Closure: Exact Coincidences, Global Existence, and Minimal Local Repair.

The task can remain compact by organizing the work into six packages rather than dozens of isolated plan points.

1. Exact coincidence classification. Classify precisely when

$$\Phi_n(\theta) = \Phi_m(\varphi)$$

for unit n, m . Include periodic and antipodal degeneracies. This is the minimum algebraic input needed to turn the transformation-valued necessary conditions into an exact existence/sufficiency theorem. A complete arbitrary word presentation is not required.

2. Joint regularity and recovered parameter regularity. Define joint continuity of $(n, \theta) \mapsto U(n, \theta)$ on the unit-axis set times $\mathbb{R}$, and then a separate C^1 strengthening if needed. Prove what this implies for $n \mapsto \alpha(n), \beta(n)$. Do not assume continuity of the rate functions by definition. Include the negative control that jointly continuous but merely parameter-indexed families still exist whenever continuous α, β are chosen.

3. Transformation-valued existence or no-go. Combine the exact coincidence theorem with the Task-XV necessary constraints. Prove one of two outcomes: construct a global jointly regular transformation-valued family, or prove that none exists. The conditional argument of proposition 1 gives a strong candidate no-go mechanism but must be formalized rather than assumed.

4. Jointly regular central composition defect. Revisit $c(g, h)$ under the strengthened regularity. Determine whether its continuously varying $Z^\times$ part remains free, reduces to a smaller class, or is forced into a discrete residual on closed relations. Preserve the distinction between the inherited reference sign and the new regular residual.

5. Minimal local repair if global closure fails. Only if package 3 proves a global obstruction, classify the weakest repair supported by the inherited data: restricted domains of directions, local regular choices, and the exact central transition on overlaps or paths. Do not name a conventional bundle, covering, connection or projective structure in the reconstruction layer. The purpose is to discover the minimal additional relational datum, not to import its standard name.

6. Final rate and Option-C audit. Recheck that the spatial rate λ remains independent under the stronger coherence conditions, then repeat the Option-C comparison only after the strengthened class is known. The final verdict must state whether the surviving datum is global, directional, local, path-dependent, discrete, or absent.

These six packages close the main Task-XV obligations and add only the nearest necessary negative controls. They deliberately exclude full Lorentz boosts, physical time evolution, interactions, field equations and a broad taxonomy of weaker regularity classes.

17 Why this is the right width for the next excavation

The growing formal project now has a practical constraint that mirrors the mathematical methodology. Every new task imports a large verified dependency tree, and the cost of rebuilding and

searching it grows even when the new mathematical question is narrow. One cannot indefinitely maximize both breadth and depth.

The correct response is not to abandon the dependency provenance, but to make the next question increasingly local to the frontier. The six-package Task XVI does this. It keeps all structures needed to decide the current central-coherence obstruction while refusing unrelated extensions.

If Task XVI closes positively, then the continuous central sector has an intrinsic global law and the difficulty moves again. If it closes negatively, the obstruction itself will determine what sort of local or relational object must be added next. Either outcome is more informative than starting a new physical branch before the present defect has been classified.

18 Conservation-of-Difficulty ledger

Question	Task-XV status
Can every one-axis lift be separated into reference and central parts?	Yes, uniquely.
Does continuity leave arbitrary functions of the one-axis parameter?	No. It forces exactly two real rates α, β .
Does differentiability further reduce the continuous class?	No. Every continuous lift is already differentiable.
Is there an exact finite mixed central composition law?	Yes; $c(g, h) \in Z^\times$ and associativity yields equation (7).
Can the inherited sign defect be separated from the new continuous residual?	Yes, exactly at the word level.
What is the infinitesimal generator?	$\alpha(n)1 + \beta(n)S - \frac{1}{2}J(n)$.
Does the spatial rate determine the central rates?	No; λ, α, β are independent generator coordinates.
Do arbitrary directional rate functions occur under the formal Task-XV regular-family predicate?	Yes.
Does that predicate impose continuity in the direction n ?	No; explicitly not.
Has joint (n, θ) regularity been classified?	No; principal hardening obligation.
Does transformation-valuedness constrain the rates?	Yes, necessarily $\alpha = 0$, $\beta \in \mathbb{Z} + \frac{1}{2}$, and β is antipodally odd.
Are those conditions sufficient?	Not proved.
Does a global transformation-valued regular family exist?	Open.
Is the exact coincidence relation $\Phi_n(\theta) = \Phi_m(\varphi)$ classified?	Open; required for the existence theorem.
Does the Option-C dependency shape reappear under axiswise regularity?	No; the formal class has two rates and zero cross-direction relations.
Is the Option-C comparison final under joint regularity?	No.
What is the narrowest next question?	Joint directional regularity plus exact transformation coincidences and global existence/no-go.

19 Conclusion

Part XV succeeds at its first and most important technical aim: it converts the arbitrary central residual of Part XIV into a completely classified regular one-axis object. Continuity is strong enough to force an exponential scaling rate and an S -rotation rate, and differentiability adds no hidden freedom. The finite central composition discrepancy is made explicit before differentiation, its associativity law is proved without quotienting, and its continuous residual component is separated from the inherited reference-word sign. The generator then splits cleanly into a central part $\alpha(n)1 + \beta(n)S$ and the already reconstructed spatial part $-\frac{1}{2}J(n)$.

The task’s official verdict that direction-dependent central data survive is mathematically correct for its formal definition of regular family. The same definition, however, makes the next Conservation-of-Difficulty boundary visible: regularity was imposed separately on each one-axis parameter and not jointly in the spatial direction. Arbitrary functions $\alpha(n), \beta(n)$ are therefore permitted by construction of the predicate. This means the theorem should not be read as the final answer to continuous $3 + 1$ coherence.

The stronger transformation-valued audit narrows the frontier dramatically. Equal visible transformations force the real scaling rate to vanish, force the second rate into a half-odd-integer set, and force antipodal oddness. Yet existence is not settled. The exact coincidence relation among visible transformations is still missing, and directional continuity has not been combined with these discrete constraints. A simple conditional argument already suggests that the jointly continuous global problem may be obstructed altogether.

The difficulty has therefore moved again, but it has also become sharply localized. It is no longer “find another algebra”, “find another spatial direction”, or “choose a better carrier”. It is now:

Can the internal central representative be chosen jointly and globally coherently over all directions?	(20)
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If yes, Task XVI should derive its exact law. If no, Task XVI should identify the minimal local or relational datum required to repair the obstruction. Only after that boundary is closed should the experiment move on to a genuinely broader dynamical or interaction question.

In the heuristic language, the old Option-C pair – common density scaling and common transport rate – has survived the one-axis classification but not yet faced the strongest multi-axis regularity test. The theory has not supplied a physical interpretation of α or β . What it has supplied is a much more precise question about them. The remaining uncertainty is no longer broad enough to hide anywhere in the spatial algebra; it is concentrated in the continuous global coherence of the central sector itself.

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